

Bernard Vulaj
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11/3/2025

Hiring Committee University of Michigan – College of Engineering
Robotics Department

Dear Hiring Committee,

I'm excited to apply for the software development position with the Robotics Department at the University of Michigan. I recently graduated with a Bachelor of Science in Computer Engineering from the Polytechnic University of Tirana, where I built a strong foundation in programming, data systems, and automation through hands-on coursework and team-based projects.

During my studies, I worked on several projects that taught me how to design, build, and refine software solutions from the ground up. I developed a React Native farm management app with real-time data handling, helped create web applications using PHP, JavaScript, and MySQL, and built database systems in Oracle to improve data accuracy and automate tasks. These experiences taught me how to take an idea, understand what users actually need, and turn it into something reliable and easy to use.

I really enjoy solving problems and improving the way things work. I'm the kind of person who likes understanding the "why" behind a process so I can help make it faster, cleaner, or easier to use. I'm comfortable learning new tools and adapting to different environments, and I take pride in being dependable and following through on what I commit to. I also really enjoy working with others. I ask questions, share ideas, and try to make sure we're all moving in the same direction.

What draws me most to Michigan Engineering is the focus on collaboration and innovation for the common good. The chance to contribute to a team that values thoughtful engineering, learning, and impact would be something I'd be genuinely excited about.

Thank you for considering my application. I'd love the opportunity to bring my technical background and collaborative mindset to your team.

Sincerely,

Bernard Vulaj

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EDUCATION

Polytechnic University of Tirana

Bachelor of Science in Computer Engineering

Grade Point Average: 3.82

Tirana, Albania

October 2022 – July 2025

- **Relevant Coursework:** C Programming, Java Programming, Data Structures, Algorithms and Advanced Programming, Web Programming, Databases, Computer Architecture, Computer OS, Automation

PROJECTS

Farm Management Mobile Application

Head Developer

Tirana, Albania

March 2025 – July 2025

- Designed and structured the application based on the principles of React Native and Expo while offering solutions such as workload management and higher product exposure through a dedicated sales platform.
- Implemented Google-based authentication with Clerk and integrated Convex backend for real-time data management, while continuously testing on real Android and iOS devices to ensure optimized performance

Car Rental Website

Group Member & Developer

Tirana, Albania

December 2024 – February 2025

- Collaborated on the design and development of a web-based car rental platform using PHP, JavaScript, and MySQL on a XAMPP development environment.
- Implemented user registration with secure email verification through PHPMailer and developed a responsive user interface for an improved booking experience.

Parking Network Management Database System

Head Developer

Tirana, Albania

April 2025 - May 2025

- Designed and implemented an Oracle-based database system to manage multi-site parking operations, ensuring efficient data organization and transaction handling.
- Created triggers and stored procedures to prevent invalid parking actions and automate payment and cancellation processes.

Calorie and Expense Tracking Website

Agile Team Developer

Tirana, Albania

January 2025

- Built a full-stack web application using Spring Boot for calorie tracking and expenditure analysis with secure authentication via Spring Security.
- Collaborated in Agile sprints, assisting with task assignment and merging completed work after each iteration to ensure smooth integration and project consistency.

Automated Palletizing System

Head Developer

Tirana, Albania

December 2023 – January 2024

- Developed a PLC-based automation simulation using WinSPS-S7 for Ladder Diagram logic to automate pallet stacking operations.
- Integrated with ITS PLC for real-time visualization and validation, optimizing logic for precise sequencing and performance efficiency.

SKILLS & ATTRIBUTES

Languages: English, Albanian and Serbian

Technical Skills: Java, C, HTML, CSS, PHP, JavaScript/TypeScript, SQL, React/React Native, Expo, Spring Boot, Git, GitHub, Convex Database, Clerk Authentication, WinSPS-S7, ITS PLC, Factory IO

Attributes: Problem-Solver, Collaborative, Detail-Oriented, Versatile, Resourceful, Adaptive